

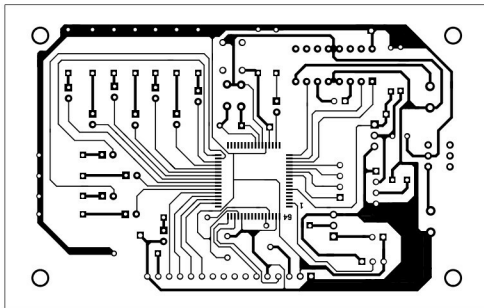


Fig. 5: An actual-size PCB of the circuit

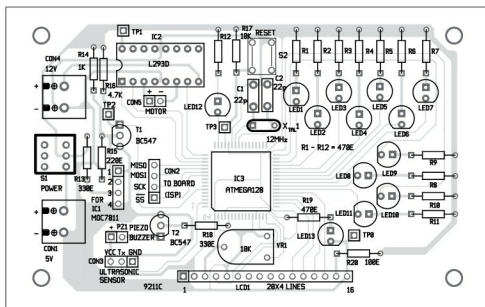


Fig. 6: Component layout of the PCB

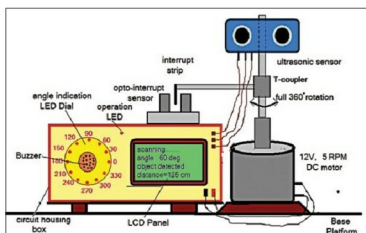


Fig. 7: Proposed system set-up

5rpm, so we can presume that in one minute (60 seconds) it completes

it has to run for one second only. Further, to reduce the speed of the motor

to 2rpm only, after moving the motor by 30° for one second, the motor rests for 1.5 seconds and then moves in a clockwise or anti-clockwise direction, depending on the direction of previous rotation.

3. As the motor rotates in clockwise or anti-clockwise direction, messages such as Scanning....., Angle:30, Angle:60, Angle:90, etc, are displayed on the LCD.

4. When the motor rotates completely by 360° in 30 seconds, the strip again passes between the gap of the slotted opto isolator module. This generates an interrupt signal and the motor starts rotating in 30° steps at a speed of 2rpm in reverse direction, and messages are displayed on the LCD as Scanning....., Angle:330, Angle:300, Angle:270, etc.

5. Same thing happens when the motor reaches back to 0°. Thus, the cycle continues and the motor rotates clockwise and anti-clockwise continuously to scan a complete 360° at the rate of 2rpm.

6. The sensor module continuously sends ultrasonic waves of 40kHz and receives echos. If it receives an echo from any object, it calculates the distance and sends it serially to the microcontroller, which displays the message Object Detected, and the distance of the object as distance = xxxcm on the LCD. The angle at which the object is detected is indicated approximately by glowing LED. It also generates short beep sounds on the buzzer for audio indication.

Construction and testing

An actual-size, single-side PCB of the circuit is shown in Fig. 5 and its component layout in Fig. 6. The slotted opto isolator module, ultrasonic sensor and DC motor are external to the PCB. These are connected to the PCB through connectors provided on the PCB.

Make a separate circular dial. Refer the proposed system set-up as shown in Fig. 7 for fitting the LEDs (LED1 through LED12) and connect these through wires from the PCB.